Computer Vision (Spring 2021) Problem Set #1

Divyesh Baskaran
dbaskaran6@gatech.edu

1a. Interesting Images

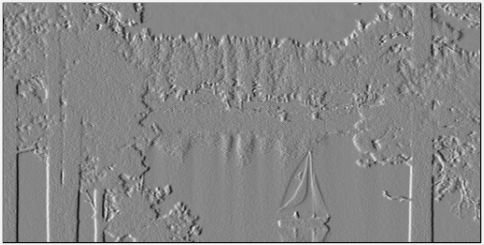


Image 1 - ps1-1-a-1



Image 2 - ps1-1-a-2

4d: Difference Image



ps1-4-d-1

5a: Noisy Green Channel



ps1-5-a-1

5b: Noisy Blue Channel



ps1-5-b-1

6a. Discussion

Between all color channels, which channel, in your opinion, most resembles a gray-scale conversion of the original. Why do you think this? Does it matter for each respective image? (For this problem, you will have to read a bit on how the eye works/cameras to discover which channel is more prevalent and widely used)

I. Between all color channels, the green channel most resembles a gray-scale conversion of the original image.

—

II. This is because the human eye is more sensitive to green light compared to red and blue light.
The luminance component of an image, which contributes to perceived brightness, is heavily influenced by the green channel.
As a result, when converting an image to grayscale, the green channel often provides a more accurate representation of the original scene's brightness and detail.

—

III. This matters since green channel has the smallest difference in pixel values between any of the images usually.

Cameras have have double the number of green pixels compared to red and blue pixels, which is why the green channel is more prevalent and widely used in image processing.

6b. Discussion

What does it mean when an image has negative pixel values stored? Why is it important to maintain negative pixel values?

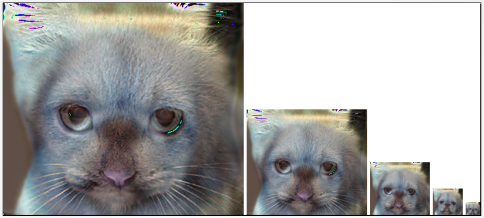
B. A negative pixel value in an image indicated the pixel is actually derived from a mathematical operation.
This could happen from an operation such as: img1 - img2, where the pixel value in img2 is greater than img1.
This could represent the difference in pixel between the two images.
—
I. This is important to preserve these negative pixel values because they can be used to represent the differene in the pixel between two images.
They should not be clipped to 0 because that would result in the destruction of information.
If destroyed, the information if it is slightly negative vs large negative could be lost.
This could be important in finding the difference in frequency of these negative vs positive pixel values.

6c. Discussion

In question 5, noise was added to the green channel and also to the blue channel. Which looks better to you? Why? What sigma was used to detect any discernible difference?

I. The Green channel looks more noisy to me.
—
II. This is because the human eye is more sensitive to green light than blue light.
The human eye has more green cones than blue cones, which means that we are better at detecting variations in green light.
As a result, noise in the green channel is more noticeable to us than noise in the blue channel.
Additionally, cameras are often designed to be more sensitive to green light, which can also contribute to the perception of more noise in the green channel.
Camera sensors often have double the number of green pixels compared to red and blue pixels, which can also make noise in the green channel more noticeable.
—
III. The lowest sigma values to notice any discernible difference for me was 8.
Any lower the noise in the green channel was very minute and hard to notice.
However, with sigma values that are greater, the noise in green channel was much easier to notice as the image got grainy.

7a: Hybrid Images



ps1-7-a-1

7b. Hybrid Images

Explain how the cutoff-frequency impacts the final hybrid image

The lower the cutoff frequency, the more low-frequency information is preserved in the hybrid image.
This means that the resulting hybrid image will have more of the original image's overall structure and less of the high-frequency details.
Conversely, a higher cutoff frequency will allow more high-frequency information to pass through, resulting in a hybrid image that retains more of the fine details from the original images.
For example, if the cutoff frequency is set to a low value, the hybrid image will appear more blurred and less detailed, while a higher cutoff frequency will produce a sharper and more detailed hybrid image.
In this case, if the cutoff frequency is set to 2 (low value), the image will look more like the dog.
The image will have less sharper details but will have more of the structure of the orignal dog image.
If the cutoff frequency is set to a higher value of 20, the resulting hybrid image will have more of the finer sharp details.
The resulting image will look more like the cat image with a lot of the sharp details from the original cat image.